AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

MAY 2014

Physics Paper II

Time: 2 hours 25 minutes Marks: 40

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are NINE questions. Answer ALL questions. Questions 7, 8 & 9 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

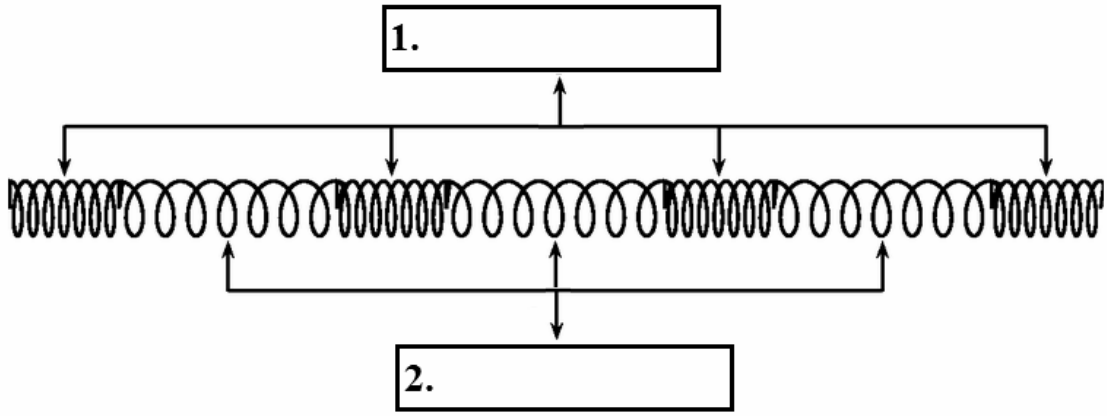
   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

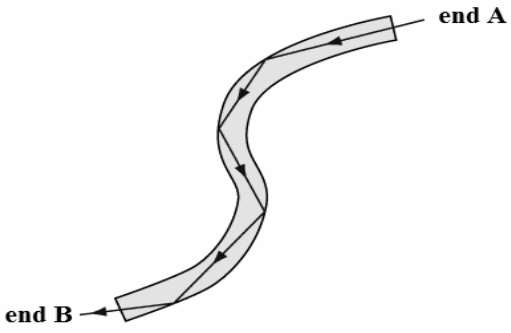
Q.1. (Total 4 Marks)

The given diagram shows a wave travelling along a slinky spring. Identify and define the type of wave travelling in the spring. Also name the regions in the given spaces.



Q.2. (Total 5 Marks)

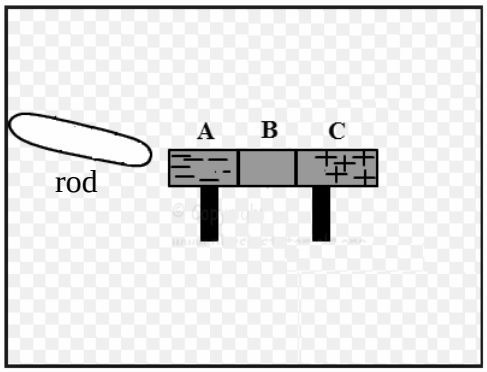
a. Briefly describe the working of an optical fibre with the help of the given diagram. (3 Marks)



b. Describe the relation between critical angle and refractive index of a substance. (2 Marks)

Q.3. (Total 6 Marks)

a. In the given figure, write the charges on the rod and describe why region **B** shows fewer charges. Name the phenomenon involved in it. (3 Marks)



b. State Ohm's law and write the value of **R** in the given table. (3 Marks)

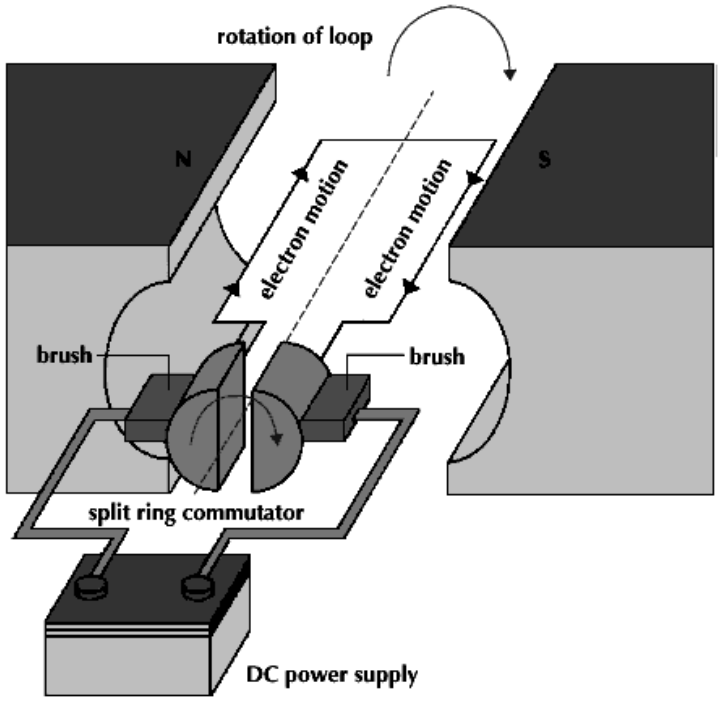
Current (Ampere)	Voltage (Volt)	Resistance (Ohm)
0.5	2.0	
1.0	4.0	
1.5	6.0	
2.0	8.0	

Q.4. (Total 3 Marks)

i. State the rule used to find the direction of current in a DC motor.

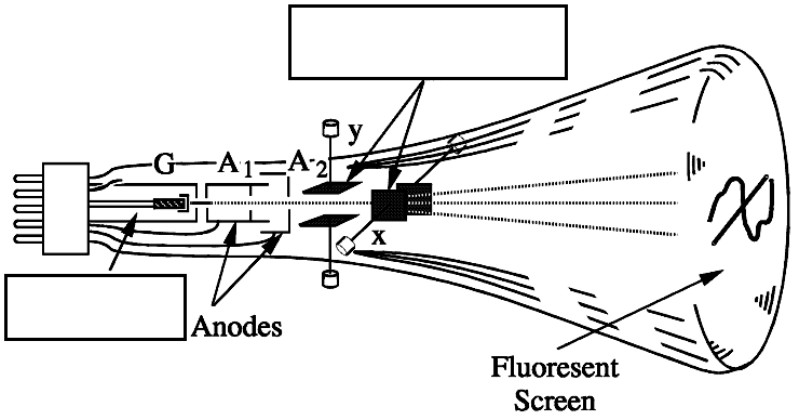
ii. What is the purpose of using a split ring commutator?

iii. Mention any ONE way for increasing the speed of a DC motor.



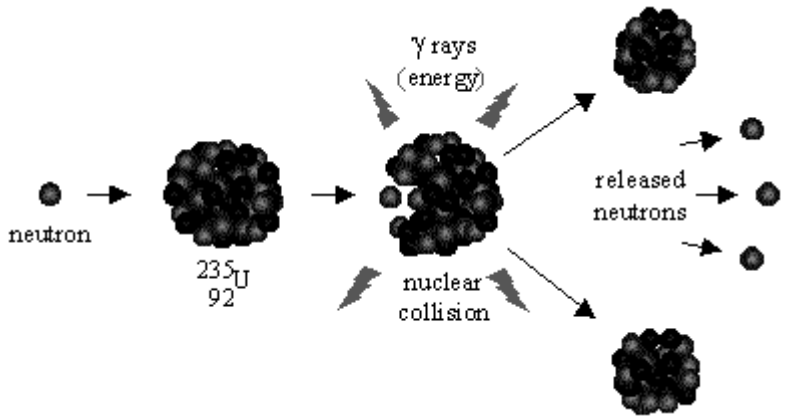
Q.5. (Total 3 Marks)

An electron gun is shown in the given diagram. Label the missing parts and write any ONE use of the electron gun.



Q.6. (Total 4 Marks)

a. Name the reaction shown in the given diagram. (2 Marks)



b. Complete the given equation for the above reaction. (2 Marks)

$$^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{\square}_{\square}Ba + {}^{\square}_{\square}Kr + \square\,{}^{1}_{0}n + E$$

Q.7. (Total 5 Marks)

EITHER

a. How can the waves in a ripple tank be made to travel more slowly and when this happens, what will be its effect on frequency and wavelength of the waves?

OR

b. Describe any FIVE uses of ultrasound in our daily lives.

Q.8. (Total 5 Marks)

EITHER

a. Draw a labelled ray diagram of a compound microscope.

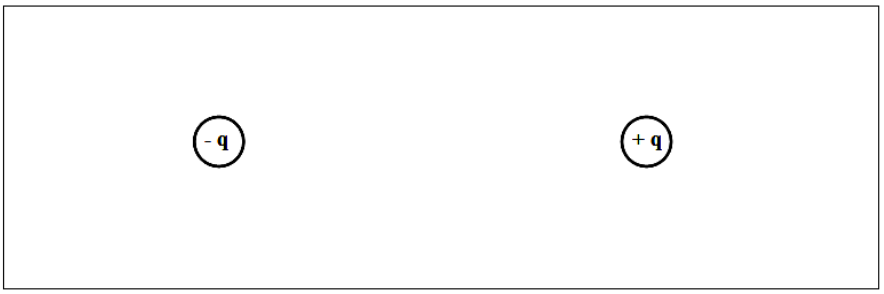
OR

b. Draw a labelled ray diagram of a telescope.

Q.9. (Total 5 Marks)

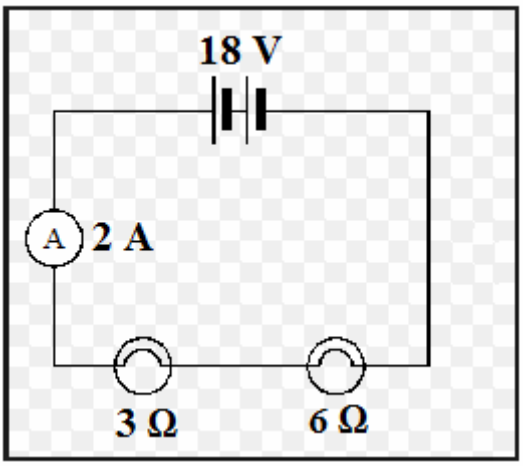
EITHER

a. Complete the given diagram of electric field surrounded by charges and describe how a glass rod is charged if rubbed with a silk cloth.



OR

b. In the given circuit diagram, bulbs are joined in series. Re-draw the diagram to show parallel combination of the bulbs. Calculate the effective resistance of both circuits by using information given in the diagram.



END OF PAPER

Please use this page for rough work